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MA3521 — $\mathbf{y}' = A\mathbf{y} + \mathbf{f}(t)$ Exponential and Trig (I) Key Name:

1. Solve $\mathbf{y}' = A\mathbf{y} + e^{-2t}\begin{pmatrix}-1\\-5\end{pmatrix}$, $\mathbf{y}(0) = \begin{pmatrix}2\\3\end{pmatrix}$ and $A = \begin{pmatrix}1 & -2\\2 & 1\end{pmatrix}$ with epairs $1+2i$, $\begin{pmatrix}1\\-i\end{pmatrix}$; $1-2i$, $\begin{pmatrix}1\\i\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$ $\begin{pmatrix}e^t\cos(2t) & -e^t\sin(2t)\\ e^t\sin(2t) & e^t\cos(2t)\end{pmatrix}\mathbf{c}$

$\alpha = 1, \quad \beta = 2$

$\operatorname{re}\mathbf{v} = \begin{pmatrix}1\\0\end{pmatrix}, \quad \operatorname{im}\mathbf{v} = \begin{pmatrix}0\\-1\end{pmatrix}$

$\operatorname{re}\left(e^{(1+2i)t}\mathbf{v}\right) = e^t\begin{pmatrix}\cos(2t)\\ \sin(2t)\end{pmatrix}$

$\operatorname{im}\left(e^{(1+2i)t}\mathbf{v}\right) = e^t\begin{pmatrix}\sin(2t)\\ -\cos(2t)\end{pmatrix}$

(b) $\mathbf{y}_p(t) =$ $e^{-2t}\mathbf{a}$ is a suitable UC form.

(c) Row reducing $\left[\,-2I - A \,\middle|\, \begin{matrix}-1\\-5\end{matrix}\,\right]$ gives the UCs in

$\mathbf{y}_p(t) =$ $e^{-2t}\begin{pmatrix}1\\1\end{pmatrix}$

$(-2I - A)\mathbf{a} = \begin{pmatrix}-1\\-5\end{pmatrix}$

$-2I - A = \begin{pmatrix}-3 & 2\\-2 & -3\end{pmatrix}$

$\mathbf{a} = \begin{pmatrix}1\\1\end{pmatrix}$

(d) Row reducing $\left[\begin{array}{cc|c}1 & 0 & 1\\0 & 1 & 2\end{array}\right]$

$\left[\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} \mid \mathbf{y}(0) - \mathbf{a}\right]$

$\begin{pmatrix}2\\3\end{pmatrix} - \mathbf{a} = \begin{pmatrix}1\\2\end{pmatrix}$

$\mathbf{c} = \begin{pmatrix}1\\2\end{pmatrix}$

gives constants $\mathbf{c}$ in

$\mathbf{y}(t) =$ $e^t\begin{pmatrix}\cos(2t) - 2\sin(2t)\\ \sin(2t) + 2\cos(2t)\end{pmatrix} + e^{-2t}\begin{pmatrix}1\\1\end{pmatrix}$

2. Solve $\mathbf{y}' = A\mathbf{y} + \cos(2t)\begin{pmatrix}1\\4\end{pmatrix}$, $\mathbf{y}(0) = \begin{pmatrix}1\\-3\end{pmatrix}$ and $A = \begin{pmatrix}1 & -2\\ 2 & 1\end{pmatrix}$ with epairs $1+2i$, $\begin{pmatrix}1\\-i\end{pmatrix}$; $1-2i$, $\begin{pmatrix}1\\i\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$ $\begin{pmatrix} e^t\cos(2t) & -e^t\sin(2t) \\ e^t\sin(2t) & e^t\cos(2t)\end{pmatrix}\mathbf{c}$

$\alpha = 1, \quad \beta = 2$

$\operatorname{re}\mathbf{v} = \begin{pmatrix}1\\0\end{pmatrix}, \quad \operatorname{im}\mathbf{v} = \begin{pmatrix}0\\-1\end{pmatrix}$

$\operatorname{re}\left(e^{(1+2i)t}\mathbf{v}\right) = e^t\begin{pmatrix}\cos(2t)\\ \sin(2t)\end{pmatrix}$

$\operatorname{im}\left(e^{(1+2i)t}\mathbf{v}\right) = e^t\begin{pmatrix}\sin(2t)\\ -\cos(2t)\end{pmatrix}$

(b) $\mathbf{y}_p(t) =$ $\mathbf{a}\cos(2t) + \mathbf{b}\sin(2t)$ is a suitable UC form.

(c) Row reducing $\left[\begin{array}{cc|c} -A & 2I & \begin{matrix}1\\4\end{matrix} \\ -2I & -A & \begin{matrix}0\\0\end{matrix}\end{array}\right]$ gives the UCs in

$\begin{pmatrix}-A & 2I\\ -2I & -A\end{pmatrix}\begin{pmatrix}\mathbf{a}\\ \mathbf{b}\end{pmatrix} = \begin{pmatrix}1\\4\\0\\0\end{pmatrix}$

$\mathbf{y}_p(t) =$ $\begin{pmatrix}-1\\-2\end{pmatrix}\cos(2t) + \begin{pmatrix}2\\0\end{pmatrix}\sin(2t)$

$= \begin{pmatrix}-1 & 2 & 2 & 0\\ -2 & -1 & 0 & 2\\ -2 & 0 & -1 & 2\\ 0 & -2 & -2 & -1\end{pmatrix}\begin{pmatrix}\mathbf{a}\\ \mathbf{b}\end{pmatrix}$

$\mathbf{a} = \begin{pmatrix}-1\\-2\end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix}2\\0\end{pmatrix}$

(d) Row reducing $\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1\end{array}\right]$

$\left[\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \mid \mathbf{y}(0) - \mathbf{a}\right]$

$\begin{pmatrix}1\\-3\end{pmatrix} - \mathbf{a} = \begin{pmatrix}2\\-1\end{pmatrix}$

$\mathbf{c} = \begin{pmatrix}2\\-1\end{pmatrix}$

gives constants $\mathbf{c}$ in

$\mathbf{y}(t) =$ $e^t\begin{pmatrix}2\cos(2t) + \sin(2t)\\ 2\sin(2t) - \cos(2t)\end{pmatrix} + \mathbf{y}_p(t)$